Biochemical and Biophysical Research Communications 836 (2026) 154567

Contents lists available at ScienceDirect

Biochemical and Biophysical Research Communications

journal homepage: www.elsevier.com/locate/ybbrc

NeoGen-BC: A synergistic framework combining generative protein language models and multi-window deep learning for designing shared neoantigens in breast cancer

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ARTICLE INFO

Keywords:
Neoantigen vaccine design
Protein language models
Generative AI
breast cancer immunotherapy
Deep learning

ABSTRACT

Breast cancer, particularly hormone receptor-positive (HR^+) and triple-negative (TNBC) subtypes, is often immunologically "cold," limiting immunotherapy efficacy. Neoantigen-based vaccines hold promise but face challenges from inaccurate identification and overreliance on HLA binding affinity, which poorly captures true immunogenicity. Most approaches focus on patient-specific (private) neoantigens, hindering scalability. In contrast, recurrent driver mutations in ESR1 and PIK3CA produce shared (public) neoantigens suitable for off-the-shelf vaccines, yet systematic discovery and design frameworks for breast cancer are lacking.

We developed NeoGen-BC, a synergistic framework advancing from neoantigen prediction to rational design. It integrates protein language model (PLM) embeddings with Retrieval-Augmented Generation (RAG) and a multi-scale MCNN-BiLSTM classifier to capture immunogenic features beyond peptide-HLA binding. A sliding-window strategy maps epitopes from driver mutations, while a controlled ProtGPT2 generative module explores immunogenic peptide space. Candidates are refined via biophysical filtering, structural validation, and immunogenicity screening.

NeoGen-BC achieved an AUC of 0.9053 on an independent test set, outperforming other machine learning models. It identified immunogenic peptides difficult to assess by MHC-II binding tools like NetMHCIIpan 4.3 and accurately detected validated MHC class II-restricted shared neoantigens from ESR1 (Y537S, D538G) and PIK3CA (H1047R) mutations, often missed by conventional predictors. De novo generated peptides matched experimentally validated ESR1 neoantigens in physicochemical and structural properties, showing favorable MHC-II binding and immunogenicity.

NeoGen-BC provides a computational foundation for next-generation "tunable antigen" vaccine design by prioritizing immunogenicity alongside peptide-MHC binding characteristics. The framework enables scalable identification of candidate peptide vaccines targeting shared breast cancer vulnerabilities and may complement emerging therapeutic strategies, including oral SERD-based approaches aimed at enhancing immune modulation and advancing precision immunotherapy.

1. Introduction

Breast cancer remains the most prevalent malignancy among women worldwide and exhibits profound molecular and clinical heterogeneity [1–5]. Hormone receptor-positive (HR^+) breast cancer, which accounts for approximately 70% of cases, is primarily driven by estrogen receptor signaling and has benefited from endocrine therapies [6–9]; however, therapeutic resistance inevitably emerges, often through mutations in ESR1 or downstream signaling pathways. Triple-negative breast cancer (TNBC), in contrast, lacks estrogen receptor, progesterone receptor, and HER2 expression, and is associated with aggressive clinical behavior, limited targeted treatment options, and poor prognosis [10–13].

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https://doi.org/10.1016/j.bbrc.2026.154567
Received 21 March 2026; Received in revised form 18 May 2026; Accepted 7 September 2026
Available online 9 September 2026
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Although immune checkpoint blockade has demonstrated efficacy in a subset of TNBC patients, durable responses remain restricted to a minority, underscoring the need for complementary immunotherapeutic strategies. Notably, recent evidence suggests that next-generation oral selective estrogen receptor degraders (SERDs) [14–17] can modulate the tumor immune microenvironment by activating interferon-γ-related pathways, enhancing antigen presentation, and increasing T cell infiltration. These immunomodulatory effects may create a therapeutic window in which cancer vaccines can be synergistically deployed to amplify antitumor immunity, particularly in HR^+ breast cancer where immunogenicity has historically been limited.

Cancer vaccines have traditionally focused on tumor-associated antigens (TAAs) [18,19], which are derived from self-proteins that are overexpressed or aberrantly expressed in tumors (Fig. 1). While TAAs offer broad applicability, central and peripheral immune tolerance often blunt their immunogenicity, resulting in weak and transient clinical responses. In contrast, tumor-specific antigens (TSAs), also referred to as neoantigens, arise from somatic mutations and are absent from normal tissues, making them highly immunogenic and ideal targets for T cell-mediated immunity [20–22]. However, most neoantigen-based vaccine strategies rely on personalized designs tailored to individual patients [23–25], which poses substantial logistical, temporal, and economic barriers to large-scale clinical deployment. To overcome these limitations, increasing attention has been directed toward shared neoantigens (also known as public neoantigens), which originate from recurrent driver mutations and are present across subsets of patients. Targeting shared neoantigens offers a compelling balance between immunogenic specificity and translational scalability, enabling the development of "off-the-shelf" vaccines that retain tumor specificity while being applicable to broader patient populations [26–28].

From a computational perspective, most existing neoantigen discovery pipelines are fundamentally discriminative: they evaluate whether a given peptide is likely to be immunogenic, bind to major histocompatibility complex (MHC) molecules, or elicit T cell responses [29–35]. While such discriminator models function as robust "judges," assessing candidate peptides derived from tumor genomes, they remain constrained by the input peptide space defined by naturally occurring mutations. This paradigm limits the exploration of potentially superior antigenic sequences that do not explicitly exist in tumor genomes but may nonetheless elicit strong and specific immune responses.

Building on our previous success with a discriminative model [36], NeoGen-BC introduces a paradigm shift that advances neoantigen vaccine design by transitioning from discrimination to generation. To ensure robustness and generalizability, our models are trained on pan-cancer peptide datasets and rigorously validated using breast cancer-specific cohorts. By integrating RAG-enhanced protein language models (PLMs) with multi-window deep learning evaluation, NeoGen-BC functions not only as a judge but also as a creator, enabling the de novo design of neoantigen peptides with highly immunological, structural, and physicochemical properties. Our discriminator module achieves strong predictive performance, reaching an AUC of 0.9053 on an independent test set, and demonstrates high accuracy in identifying shared neoantigen peptides presented by hormone receptor-positive (HR^+) and triple-negative breast cancer (TNBC) tumors. Leveraging this predictive capability, the generative module produces structurally stable and biochemically plausible peptides with favorable HLA binding potential, which are subsequently verified by the NeoGen-BC discriminator. We focus specifically on shared neoantigens derived from recurrent breast cancer driver genes, using ESR1 mutations for de novo peptide generation due to their prevalence in hormone receptor-positive (HR^+) breast cancer and clinical relevance.

Collectively, NeoGen-BC establishes a new computational paradigm for scalable neoantigen vaccine development, bridging generative artificial intelligence with cancer immunology and providing a systematic foundation for prioritizing candidate peptides for subsequent in vitro and translational validation.

2. Materials and methods

Fig. 2 presents an overview of the NeoGen-BC workflow. Stage A involves the curation of experimentally validated epitope data from the CEDAR database, which are subsequently partitioned for five-fold cross-validation and independent internal testing. Stage B employs the proposed MCNN-BiLSTM architecture to discriminate neoantigen epitopes from self-derived antigenic peptides by estimating their immunogenic probability. Stage C leverages a subset of the CEDAR corpus to construct a retrieval-augmented generation (RAG) module, providing contextual biological knowledge to enhance epitope representation learning during model training. Stage D integrates a generative protein language model to design de novo peptide candidates, which are subjected to a series of computational and biochemical filtering steps. Stage E shows the final evaluation by the NeoGen-BC predictor for these filtered sequences. Stage F presents the generated peptide's visualization, supported by computational tools.

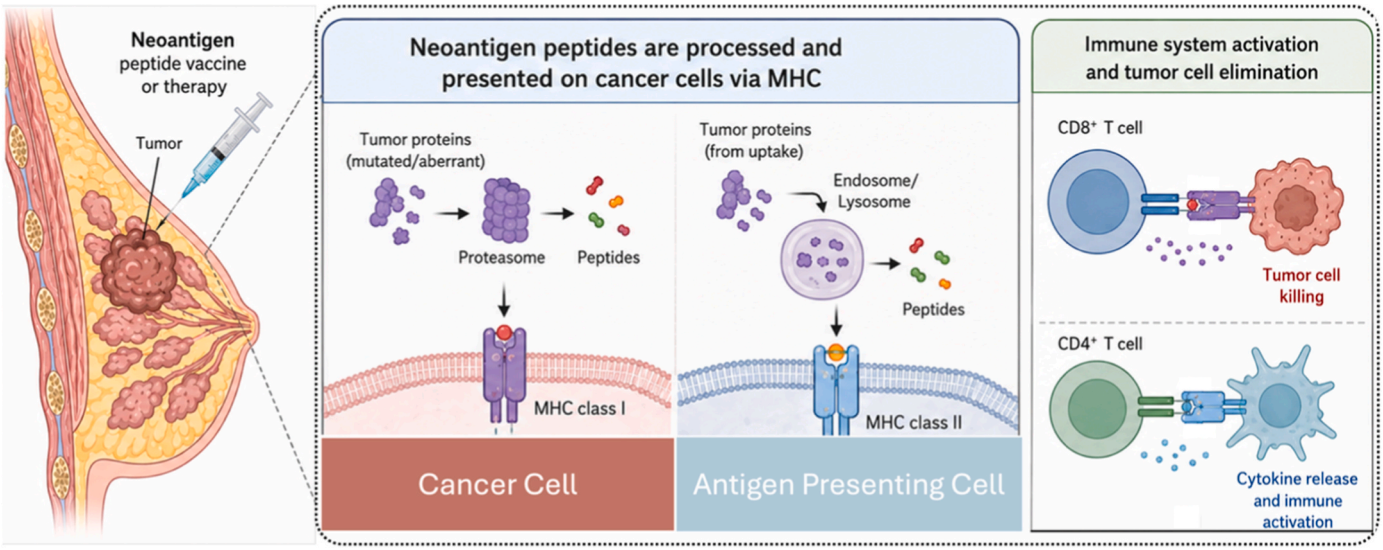


Fig. 1. The mechanism of neoantigen peptide vaccine in treatment of breast cancer.

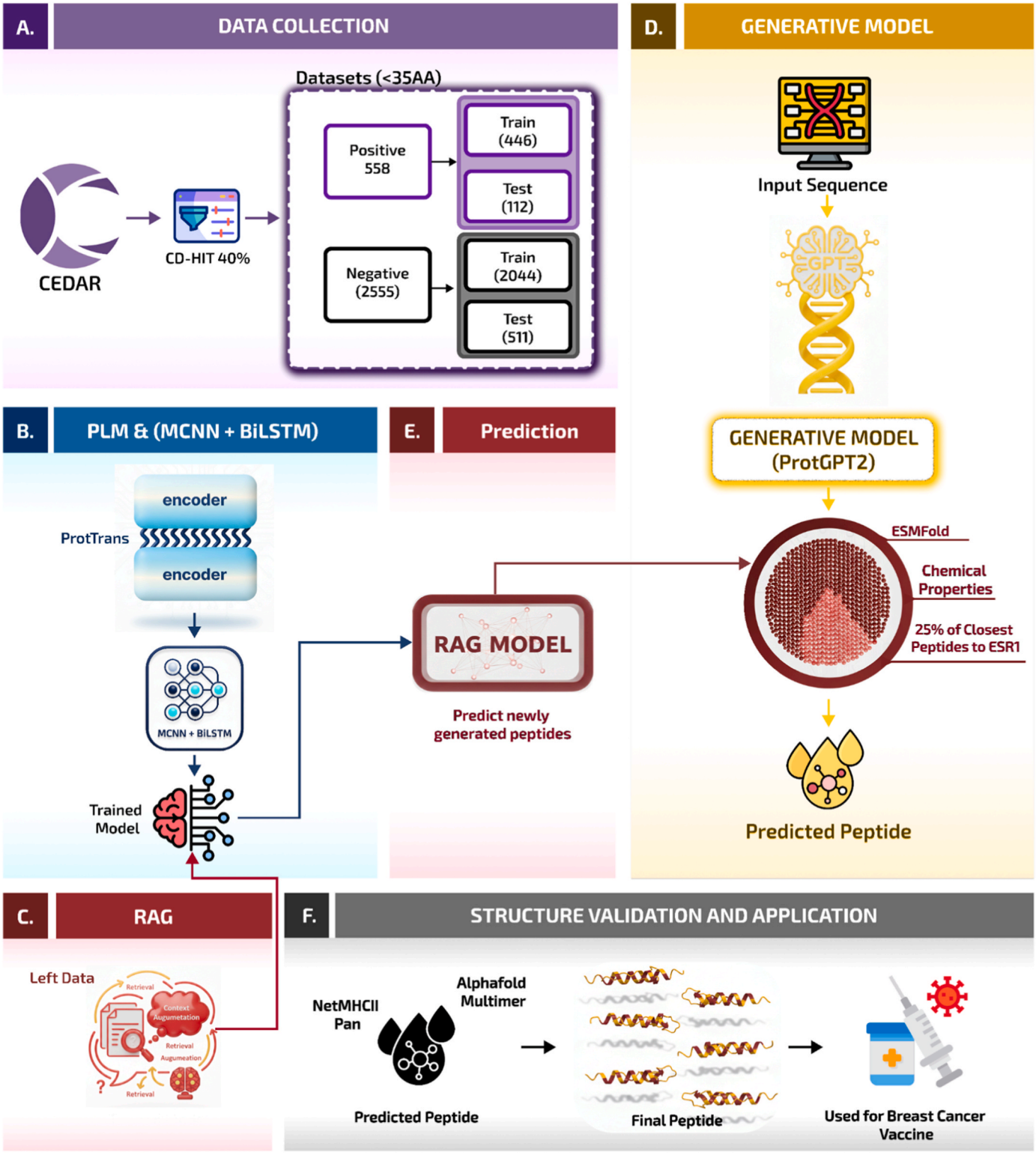


Fig. 2. Computational pipeline for generating and evaluating neoantigen peptides.

2.1. Data collection

In this study, experimentally validated epitope data were collected from the Cancer Epitope Database and Analysis Resource (CEDAR) at https://cedar.iedb.org. In contrast to our previous melanoma-focused work, we adopted a pan-cancer data acquisition strategy to support broader generalization. Specifically, neoantigen peptides were retrieved using the keywords "Linear Peptide", "Neoantigen", and "Any" cancer type. To construct a representative set of non-neoantigen controls, peptides annotated as Self/Host Antigen were collected after explicitly excluding entries labelled as neoantigens.

This initial curation resulted in 6516 neoantigen peptides and 377,990 self-derived antigen peptides. To reduce redundancy and mitigate sequence homology bias, all peptide sequences were clustered

using CD-HIT with a sequence identity threshold of 40%. After homology filtering, 561 neoantigen peptides and 2823 self-antigen peptides remained. Subsequently, only peptides with lengths shorter than 35 amino acids were retained to eliminate outliers and noise, leaving 558 neoantigen peptides and 2555 other self-antigen peptides. This length constraint is consistent with the intrinsic characteristics of the CEDAR dataset and reflects the typical length distribution of MHC-presented epitopes.

The curated dataset was then randomly partitioned into training (80%) and independent testing (20%) sets. The training set comprised 446 neoantigen peptides and 2044 self-antigen peptides, while the testing set included 112 neoantigen peptides and 511 self-antigen peptides. This strict separation between training and testing data was designed to ensure robust performance evaluation and to assess the generalizability of the proposed deep learning model.

To construct the RAG knowledge base, we utilized the remaining CEDAR-derived peptides prior to the final training-testing split. This corpus consisted of 5958 neoantigen peptides and 375,435 self-antigen peptides. CD-HIT was again applied with a 40% sequence identity threshold to remove highly homologous sequences, thereby minimizing redundancy and optimizing storage efficiency while preserving epitope diversity within the RAG database. The resulting retrieval dataset contained 705 neoantigens and 3579 other antigens, which we designated as the RAG_I database. To further evaluate the robustness of the RAG framework under stricter sequence separation conditions, we additionally constructed a homology-purged retrieval database by removing all sequences exhibiting greater than 40% similarity to peptides in the independent test set. This filtering process resulted in a reduced dataset containing 462 neoantigens and 2636 other antigens. The resulting homology-purged database was used to assess whether the performance improvements of the RAG framework persisted after minimizing potential sequence similarity bias. This second retrieval database is referred to as RAG_II.

In addition to the CEDAR-derived data, we curated an external validation dataset from published studies, consisting of 25 experimentally reported shared neoantigens and 49 tumor-associated antigens [27]. This independent dataset was used exclusively for external evaluation to further assess the predictive robustness and translational relevance of NeoGen-BC beyond the training domain.

Tables 1 and 2 summarize the survey datasets in our model training and validation steps.

2.2. Feature extraction

In this study, peptide sequences were encoded using embeddings derived from protein language models (PLMs), which have demonstrated strong capability in capturing contextual, structural, and evolutionary information from amino acid sequences. Our primary feature representation was obtained from ProtTrans [37], and its performance was systematically compared with two widely adopted PLMs: ESM-2 [38] and TAPE [39].

ProtTrans embeddings were generated using the ProtT5-XL-UniRef50 model, which is based on a transformer architecture pretrained on large-scale protein sequence corpora using a masked language modeling objective. ProtTrans was trained on the UniRef50 dataset, comprising approximately 45 million protein sequences, enabling the model to learn rich contextual representations across diverse biological families. The ProtT5-XL model consists of 24 transformer layers and contains approximately 3 billion parameters, generating a 1024-dimensional contextual embedding vector for each amino acid residue. For simplicity, this protein language model is hereafter referred to as ProtT5 throughout this study.

Table 1
The collection of neoantigen and other self-antigen peptides from the CEDAR database.

Dataset	Epitopes	Similarity < 0.4	Length < 35
Neoantigens	6516	561	558
Others	377990	2823	2555
Dataset	**Training data (80%)**		**Independent Test (20%)**
Neoantigens	446		112
Others	2044		511

Table 2
Statistics of the RAG database.

Dataset	Origin	Remove similarity < 40%	CD-HIT-2D with Test Set
Neo antigens	5958	705	462
Other antigens	375435	3579	2636

For comparative evaluation, we additionally employed ESM-2 and TAPE embeddings. ESM-2 is a state-of-the-art protein language model developed by Meta AI, pretrained on the UniRef and UniRef90 protein sequence databases using transformer architectures scaled to millions of parameters. In this study, ESM-2 embeddings with 1280-dimensional residue-level representations (esm2_t33_650 M_UR50D variant) were used, corresponding to models with deep transformer stacks optimized for capturing long-range dependencies in protein sequences. TAPE, in contrast, represents an earlier generation of protein language models and was pretrained on the Pfam and UniProt datasets using a masked language modeling strategy. The TAPE transformer consists of 12 layers and outputs 768-dimensional residue embeddings, offering a comparatively lightweight representation.

All peptide sequences were converted into residue-level embeddings and subsequently organized into fixed-length (35 residues) matrices to serve as input to the downstream deep learning architecture. Padding or truncation was applied where necessary to ensure consistent input dimensionality across peptides.

Among the evaluated PLMs, ProtT5 embeddings consistently demonstrated superior performance in discriminating neoantigen peptides from self-derived antigen peptides. We attribute this advantage to several factors. First, ProtT5 was pretrained on a substantially larger and more diverse protein corpus, enabling enhanced modeling of biochemical context and subtle sequence variations introduced by somatic mutations. Second, the ProtT5 architecture effectively captures local and global residue dependencies, which are critical for modeling epitope immunogenicity that arises from short peptide sequences rather than full-length proteins. Finally, the embedding dimensionality of ProtT5 strikes a favorable balance between representational richness and computational efficiency, allowing the downstream multi-window convolutional neural network to effectively exploit both local motif patterns and broader contextual signals.

Based on these observations, ProtT5 embeddings were selected as the default feature representation for the NeoGen-BC framework, while ESM-2 and TAPE were retained as benchmark baselines to assess the robustness and generalizability of our approach.

2.3. The RAG module

To improve the robustness and generalization of the NeoGen-BC predictor, we incorporated a Retrieval-Augmented Generation (RAG) module into the training pipeline. Given the limited availability and imbalance of experimentally validated neoantigen data, RAG enables the model to leverage additional biologically relevant context beyond individual peptide representations. Recently, RAG has emerged as a widely adopted strategy for improving the performance and data efficiency of language models [40–42].

In NeoGen-BC, ProtT5 embeddings are used to retrieve semantically similar peptides from a curated, redundancy-reduced knowledge base derived from CEDAR. For each query peptide embedding, Euclidean distance (L2 norm) was calculated between the query ProtT5 embedding

and all embeddings stored in the retrieval database. The top-5 nearest neighbor embeddings were then retrieved based on the smallest Euclidean distances. These retrieved embeddings were subsequently combined using mean pooling to generate a contextual representation vector. The final RAG-enhanced embedding was computed as a weighted fusion between the original query embedding and the averaged contextual embedding. This retrieved context allows the model to learn from both local peptide features and broader distributional patterns within the immunopeptidome. We additionally evaluate several fusion ratios including 2:1, 1:1, and 9:1 during model development. Among these, the 9:1 ratio achieved the most balanced performance on both training and independent test datasets, whereas larger contextual contributions tended to increase overfitting on the independent test set (Fig. S2).

By enriching peptide representations with contextual information, the RAG module reduces overfitting, enhances pan-cancer generalization, and improves the identification of shared neoantigens. In our study, RAG serves as a scalable and effective strategy to augment ProtT5-based deep learning models for neoantigen prediction.

2.4. The predictive module (MCNN-BiLSTM architecture)

To accurately discriminate neoantigen peptides, NeoGen-BC adopts a hybrid MCNN-BiLSTM architecture that jointly models local amino acid motifs and their sequential dependencies using protein language model embeddings. The use of MCNN-based multi-window scanning builds upon our previous work [43–45]. Specifically, each peptide is encoded as a fixed-length matrix derived from ProtT5 embeddings (1024 features per residue). The input is processed by multiple parallel convolutional branches with different window sizes (for example, 8, 12, 16, 20, 24, 28, and 32 residues), enabling multi-scale motif extraction. Within each branch, a one-dimensional convolutional layer with 256 filters scans the sequence to capture scale-specific local patterns, while preserving residue order.

The convolutional outputs from each branch are directly fed into independent bidirectional LSTM (BiLSTM) layers with 24 hidden units, allowing the model to capture contextual dependencies by integrating information from both upstream and downstream residues. For each branch, the forward and backward hidden states are concatenated to form a 48-dimensional context-aware representation. These representations are subsequently concatenated across all branches, regularized by dropout, and passed to fully connected layers for binary classification of neoantigen versus self-antigen peptides. A schematic overview of the architecture is shown in Fig. 2.

2.5. The generative module

To enable de novo neoantigen design and systematic exploration of immunogenic sequence space, NeoGen-BC incorporates a generative protein language modeling module based on ProtGPT2 [46]. ProtGPT2 is an autoregressive transformer model adapted from the GPT-2 architecture and trained exclusively on protein sequences to learn the statistical grammar underlying amino acid composition and ordering. The model was pretrained on approximately 50 million protein sequences derived from the UniRef50 database using a causal language modeling objective, allowing it to generate biologically plausible protein fragments without explicit structural supervision. ProtGPT2 consists of 36 transformer decoder layers, employs multi-head self-attention, and contains approximately 738 million parameters, making it well suited for conditional protein sequence generation tasks.

In NeoGen-BC, ProtGPT2 is used to generate candidate peptide sequences conditioned on biologically meaningful mutation-centered contexts derived from recurrent breast cancer driver mutations, particularly in ESR1. Specifically, short amino acid segments surrounding hotspot mutations (for example, K303R, E380Q, Y537S, and D538G) were used as context prompts to anchor the generative process within relevant oncogenic sequence neighborhoods. This conditional generation strategy ensures that the produced sequences remain proximal to known mutation-associated regions while allowing controlled exploration of local sequence variations.

Sequence generation was performed using stochastic nucleus sampling (top-p sampling) combined with temperature scaling to balance diversity and biological plausibility. For each mutation context, ProtGPT2 was prompted to generate extended peptide fragments, from which overlapping 15-mer peptides were extracted using a sliding window approach [47]. Peptides containing non-standard amino acids or invalid characters were discarded. This procedure yielded a diverse pool of candidate sequences that preserved mutation-adjacent context while introducing novel amino acid combinations not explicitly observed in the training data.

The generated peptides serve two complementary purposes within the NeoGen-BC framework. First, they function as decoy sequences, enabling systematic probing of the classifier's decision boundary by introducing hard-to-classify peptides that closely resemble authentic neoantigens yet differ subtly in sequence composition. Second, they act as optimized candidates, enriching the peptide space with sequences that may exhibit enhanced immunogenic potential relative to naturally occurring mutation-derived peptides. By evaluating these generated peptides using the MCNN-BiLSTM discriminator, NeoGen-BC effectively identifies potential regions of sequence space associated with high predicted neoantigenicity.

Importantly, this generative-discriminative coupling enables NeoGen-BC to move beyond conventional neoantigen screening toward rational neoantigen design, providing a scalable computational strategy for discovering shared neoantigen candidates with optimized immunological properties. The integration of ProtGPT2 thus transforms the framework from a purely evaluative system into an active design platform for next-generation cancer vaccine development.

2.6. Performance evaluation

Model performance was assessed using a diverse set of standard evaluation metrics to provide a comprehensive and unbiased evaluation of predictive capability. These metrics included Sensitivity, Specificity, Accuracy, Matthews Correlation Coefficient (MCC), Area Under the Receiver Operating Characteristic Curve (AUC), and Area Under the Precision-Recall Curve (AUPRC). Sensitivity, also referred to as recall, quantifies the model's ability to correctly identify positive samples, whereas Specificity measures its effectiveness in correctly recognizing negative instances. Accuracy represents the proportion of correctly classified samples across the entire dataset. To account for class imbalance, MCC was additionally employed because it integrates true positives, true negatives, false positives, and false negatives into a single correlation-based metric, providing a more balanced assessment of classification quality.

Receiver Operating Characteristic (ROC) curves and Precision-Recall (PR) curves were further generated to evaluate model behavior across varying classification thresholds. While AUC summarizes the overall discriminative capability between positive and negative classes, AUPRC provides a more informative evaluation under class-imbalanced conditions (~1:1.4 in the test set) by emphasizing precision and recall performance for the positive class.

The formal mathematical definitions of these evaluation metrics are provided in the Supplementary Materials.

3. Results

3.1. Model optimization

3.1.1. Training hyperparameter tuning

To systematically optimize the predictive performance of the proposed MCNN-BiLSTM architecture, we conducted a comprehensive

hyperparameter exploration using ProtT5-derived embedding features. The optimization strategy focused on evaluating key architectural components that govern the model's capacity to capture sequence-level and contextual information, including convolutional window size, multi-window configurations, number of convolutional filters, fully connected hidden layer dimensionality, and BiLSTM unit size. Each component was assessed independently under a five-fold cross-validation framework to ensure robustness and minimize overfitting. By progressively refining these parameters, we aimed to identify an architecture that achieves an optimal balance between representational power and generalization for neoantigen prediction. The hyperparameter optimization experiments reported in Tables S1–S5 were conducted prior to RAG integration using five-fold cross validation on the training set only.

Table S1 evaluated the discriminative capability of individual convolutional window sizes. Performance generally improved as the window size increased, indicating the importance of capturing broader sequence context for neoantigen prediction. Smaller windows (e.g., 4 and 8 residues) yielded lower MCC and AUC values, suggesting limited ability to model higher-order dependencies. The best single-window performance was achieved with a window size of 32, attaining an AUC of 0.8486 and the highest MCC of 0.4879, while maintaining balanced sensitivity and specificity. These results indicate that longer receptive fields are beneficial for capturing informative sequence patterns in neoantigen peptides.

Subsequently, we explored the multi-window configurations (Table S2) to assess whether integrating multiple receptive fields could further enhance predictive performance. In general, multi-window architectures outperformed most single-window models, highlighting the advantage of combining local and global sequence features. The best-performing configuration incorporated seven window sizes (8, 12, 16, 20, 24, 28 and 32), achieving the highest AUC of 0.8494 and MCC of 0.4935, along with improved accuracy (0.8028) and specificity (0.8084). These findings demonstrate that multi-scale feature integration provides complementary information that strengthens neoantigen discrimination.

Additionally, filter size optimization in Table S3 revealed that increasing the number of convolutional filters generally improved performance up to a moderate scale. A filter size of 256 yielded the highest AUC (0.8494) and competitive MCC (0.4935), outperforming both smaller and larger configurations. While larger filter sizes (512 and 1024) showed increased sensitivity or specificity in isolation, they did not translate into consistent overall gains, suggesting diminishing returns and potential overparameterization beyond 256 filters.

In Table S4, the fully connected hidden layer analysis indicated that moderate hidden layer sizes achieved the best balance between expressiveness and generalization. A hidden layer size of 500 neurons delivered the strongest overall performance, with the highest AUC (0.8494) and a robust MCC of 0.4935. Increasing the hidden layer size beyond this point did not yield measurable improvements and in some cases resulted in reduced performance, likely due to overfitting.

We finally analyzed the BiLSTM unit tuning in Table S5. The result demonstrated that sequence modeling capacity critically influences neoantigen prediction. Among all configurations, 24 BiLSTM units achieved the best overall performance, producing the highest accuracy (0.8229) and MCC (0.5121), as well as a strong AUC of 0.8485. This configuration effectively balanced sensitivity and specificity, suggesting that moderate recurrent capacity is sufficient to capture relevant contextual dependencies without introducing excessive model complexity.

Overall, these hyperparameter analyses support the final MCNN-BiLSTM configuration used in NeoGen-BC, which integrates multi-window convolutional scanning, 256 filters, a 500-unit hidden layer, and 24 BiLSTM units, achieving robust and consistent performance across five-fold cross validation using ProtT5 embeddings.

3.1.2. Pretrained language model comparison

Table S6 compares the performance of different protein language model (PLM) embeddings (ProtT5, ESM-2, and TAPE) using the same MCNN-BiLSTM architecture and a shared set of hyperparameter configurations in Section 3.1.1. Under this controlled experimental setting, ProtT5 achieved the strongest overall performance across five-fold cross validation, with Accuracy of 0.8229, MCC of 0.5121, and AUC of 0.8485. ESM-2 and TAPE achieved comparatively lower MCC and AUC values under the same architectural configuration.

These results suggest that ProtT5 embeddings were more compatible with the current MCNN-BiLSTM framework and training configuration used in this study. One possible explanation is that ProtT5 embeddings capture rich contextual and biologically relevant sequence representations derived from large-scale protein pretraining. However, we note that the architecture and hyperparameters were primarily optimized using ProtT5 embeddings and were subsequently applied unchanged to ESM-2 and TAPE for controlled comparison. Therefore, the present results may not be interpreted as a definitive ranking of PLMs under individually optimized conditions, as alternative architectures or hyperparameter settings may further improve the performance of ESM-2 or TAPE.

Based on the observed performance within this experimental setting, ProtT5 embeddings were selected as the default feature representation for the NeoGen-BC framework in subsequent experiments.

3.2. Performance using RAG module

We explored the addition of a retrieval-augmented generation (RAG) module to the MCNN-BiLSTM framework to assess whether incorporating external epitope-related knowledge could enhance neoantigen prediction performance across different protein language model embeddings. The RAG-enhanced models were evaluated using ProtT5, ESM-2, and TAPE embeddings under both five-fold cross validation and independent testing on the pan-cancer dataset. In these experiments, the RAG_I retrieval database was constructed and applied separately for each corresponding PLM embedding space. The comparative performance results are summarized in Table 3.

Under five-fold cross validation, ProtT5 combined with RAG (ProtT5-RAG) demonstrated a marked performance improvement, achieving 0.8155 sensitivity, 0.8408 specificity, and an overall accuracy of 0.8357, together with a substantially elevated MCC of 0.5652 and an AUC of 0.8940. These results indicate that the retrieval-augmented contextual information significantly strengthens the model's ability to distinguish neoantigens from self-derived peptides. Although ESM2-RAG and TAPE-RAG also benefited from RAG integration, their gains were comparatively limited, with lower MCC and AUC values relative to ProtT5-RAG.

This performance advantage persisted on the independent test set, where ProtT5-RAG maintained strong generalization, yielding an accuracy of 0.8427, an MCC of 0.5727, and an AUC of 0.9053. Importantly, the model achieved a favorable balance between sensitivity (0.8125) and specificity (0.8493), underscoring its robustness in identifying true neoantigens while effectively controlling false positives. In contrast, the ESM2-RAG and TAPE-RAG models exhibited reduced

Table 3
Performance of using RAG-enhanced embeddings on different PLMs.

Feature	Sens	Spec	Acc	MCC	AUC
Cross-validation (5 folds)					
ProtT5-RAG	0.8155	0.8408	0.8357	0.5652	0.8940
ESM2-RAG	0.7861	0.7795	0.7803	0.4643	0.8378
TAPE-RAG	0.8128	0.7805	0.7859	0.4874	0.8516
Independent Test					
ProtT5-RAG	0.8125	0.8493	0.8427	0.5727	0.9053
ESM2-RAG	0.7679	0.7867	0.7833	0.4592	0.8511
TAPE-RAG	0.8214	0.8023	0.8058	0.5180	0.8715

generalization performance, with lower accuracy and MCC despite moderate AUC improvements compared to their non-RAG counterparts. The ROC and PR illustration for these RAG-enhanced PLM features are shown in Fig. 3.

Overall, these findings demonstrate that the integration of a RAG module provides informative supplementary knowledge during training, substantially enhancing predictive robustness, particularly when coupled with high-capacity embeddings such as ProtT5. This synergistic effect supports the inclusion of RAG as a core component of the NeoGen-BC framework.

3.3. Performance comparison with other classifiers

We further compared the fully optimized MCNN-BiLSTM framework against a range of conventional machine learning and deep learning classifiers on the independent test set to evaluate its relative effectiveness for neoantigen prediction, as summarized in Table 4 and Fig. 4. In this experiment, ProtT5 embeddings integrated with the RAG_I retrieval database were used as input features and evaluated across multiple classification architectures to assess the contribution of the proposed MCNN-BiLSTM framework. The evaluated models included MCNN without recurrent modeling, support vector machine (SVM), k-nearest neighbors (KNN), random forest (RF), extra trees (ET), a standard convolutional neural network (CNN), a multilayer perceptron (MLP), Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), all trained and tested under identical data splits.

Among all evaluated approaches, MCNN-BiLSTM achieved the best overall performance, with a sensitivity of 0.8125, specificity of 0.8493, and an accuracy of 0.8427. Notably, it produced the highest MCC (0.5727) and AUC (0.9053), indicating superior robustness and discriminative power in distinguishing neoantigen peptides from self-derived antigens. These results underscore the effectiveness of integrating bidirectional sequence modeling with multi-scale convolutional feature extraction.

Direct comparison between MCNN-BiLSTM and MCNN alone highlights the critical contribution of the BiLSTM component. While MCNN without BiLSTM achieved reasonable performance (MCC = 0.5342, AUC = 0.8979), the addition of BiLSTM substantially improved accuracy, MCC, and AUC, demonstrating that explicit modeling of long-range contextual dependencies among convolutional features is essential for accurate neoantigen recognition.

Traditional machine learning classifiers, such as SVM and ensemble-based methods (RF and ET), exhibited high specificity but consistently lower sensitivity, indicating a tendency to miss true neoantigen peptides despite achieving competitive accuracy or AUC values. In contrast, distance-based (KNN) and shallow CNN models performed poorly, particularly in MCC, reflecting limited capacity to capture complex sequence patterns inherent to immunogenic epitopes. The MLP showed comparable performance but remained inferior to MCNN-BiLSTM in both robustness and overall discrimination.

Recurrent neural network baselines, including standalone LSTM and GRU models, showed moderate predictive ability, achieving AUC values of 0.8025 and 0.8374, respectively. However, both architectures exhibited lower sensitivity and MCC compared to MCNN-BiLSTM, suggesting that sequence modeling alone is insufficient without prior multi-scale motif extraction. These findings underscore the importance of coupling recurrent architectures with convolutional feature learning to enhance neoantigen recognition.

Fig. 4 illustrates the ROC curves (Panel A) and PR curves (Panel B) comparing the MCNN-BiLSTM framework with alternative machine learning and deep learning models. The MCNN-BiLSTM model trained using RAG-enhanced ProtT5 embeddings consistently achieved superior curve profiles and larger AUC and AUPRC values relative to the other evaluated methods, indicating improved discriminative performance for neoantigen prediction.

Overall, these results demonstrate that the MCNN-BiLSTM architecture outperforms both conventional machine learning models and simpler deep learning baselines, confirming that the synergistic integration of multi-scale motif detection and bidirectional sequence context modeling is crucial for reliable neoantigen peptide classification.

3.4. Performance using homology-purged RAG knowledge base

To further justify that the model does not look up near-duplicates of test sequences, we re-run evaluation with a homology-purged RAG knowledge base (RAG_II). In this RAG_II setting, we removed all sequences with $\geq$ 40% similarity to the test set using the CD-HIT-2D tool [48]. Although performance decreased moderately relative to the original RAG configuration, with AUC values decreasing from 0.8940 to 0.8712 in five-fold cross-validation and from 0.9053 to 0.8790 on the independent test set, the homology-purged ProtT5-RAG framework continued to substantially outperform the non-RAG baseline in both MCC and AUC metrics. These findings (Table S7) suggest that the observed performance gains were not solely attributable to retrieval of near-duplicate sequences but instead reflect the contribution of biologically relevant contextual information provided by the RAG framework.

In addition, we also tested the classification models with the homology-purged RAG database to further evaluate whether the predictive performance remained robust after eliminating potential sequence overlap between the retrieval database and the independent test set. As shown in Fig. S1, MCNN-BiLSTM continued to achieve the highest PR-AUC (0.6610) together with a strong AUC value (0.8790), indicating improved positive-class discrimination under imbalanced conditions. Although MCNN achieved a slightly higher MCC, MCNN-BiLSTM demonstrated substantially higher sensitivity, suggesting superior capability for identifying true neoantigen peptides. These results emphasize that integrating bidirectional contextual sequence modeling with multi-window convolutional feature extraction provides additional discriminative information beyond convolutional motif learning alone, while also supporting the robustness of the proposed framework after leakage-controlled evaluation.

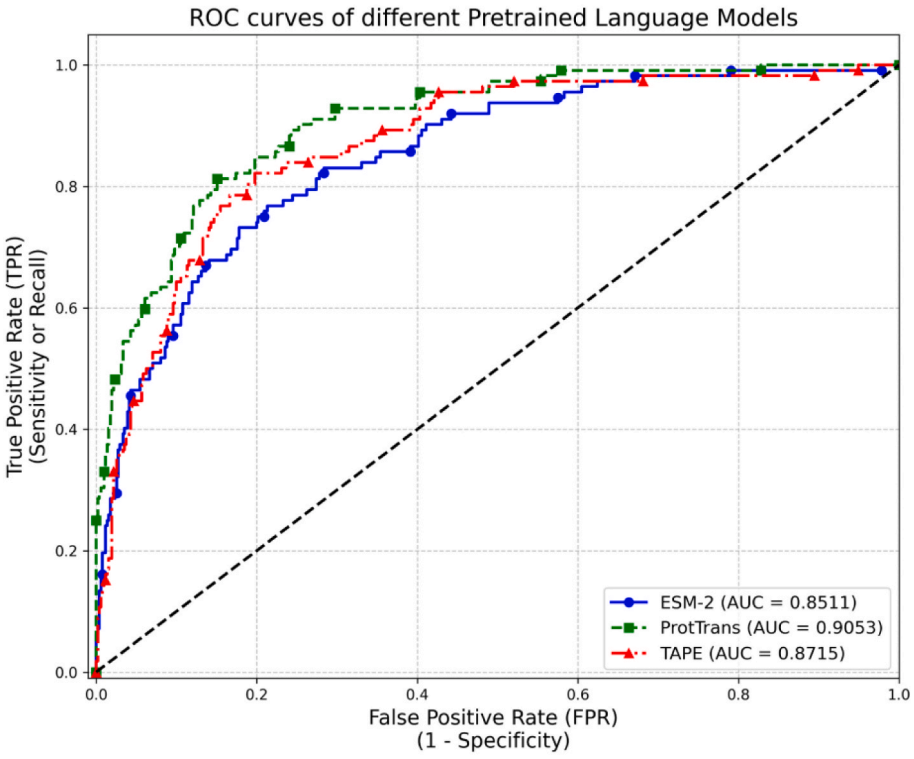
Although the homology-purged RAG_II database provided a stricter leakage-controlled evaluation setting, the original RAG_I configuration more closely reflects realistic biological retrieval scenarios in which related immunogenic peptides and partially homologous antigen contexts are naturally present within expanding epitope databases. The improved performance observed with RAG_I therefore likely captured the practical advantage of leveraging biologically relevant contextual information rather than merely memorizing identical sequences. Importantly, the retention of strong predictive performance under the stricter RAG_II setting supports that the NeoGen-BC framework maintains genuine discriminative capability even after controlling for potential sequence redundancy. Based on these observations, the subsequent case studies and downstream analyses primarily utilized the original RAG_I configuration to maximize biologically informative retrieval context while maintaining robustness validated through the homology-purged evaluation.

3.5. Identification of shared neoantigens

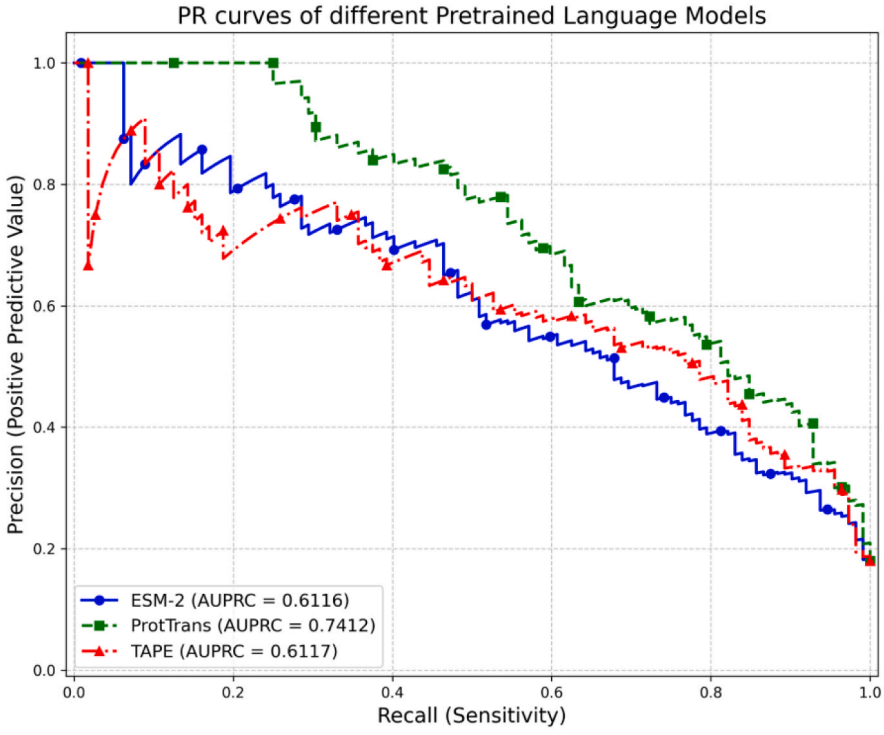
3.5.1. Pan-cancer generalization to breast cancer

To evaluate whether a model trained exclusively on pan-cancer neoantigen data can generalize effectively to breast cancer-specific immune contexts, we conducted a focused case study using experimentally validated MHC class I-associated peptides (MAPs) derived from hormone receptor-positive (HR^+) and triple-negative breast cancer (TNBC) tumors. These peptides, curated from a published immunopeptidomics study [27], include both tumor-specific antigens (TSAs; shared neoantigens) and tumor-associated self antigens (TAAs), providing a stringent and biologically relevant test set independent of model training.

As summarized in Table S8, the dataset comprised 25 shared



A



B

Fig. 3. Receiver Operating Characteristic (ROC) curves (A) and Precision-Recall (PR) curves (B) comparing the performance of different RAG-enhanced protein language model (PLM) embeddings on the independent test set across varying classification thresholds.

Table 4
Performance of MCNN-BiLSTM and other different predictors in the independent test.

Method	Sens	Spec	Acc	MCC	AUC
MCNN-BiLSTM	0.8125	0.8493	0.8427	0.5727	0.9053
MCNN	0.8304	0.8102	0.8138	0.5342	0.8979
SVM	0.5804	0.9295	0.8668	0.5313	0.9017
KNN	0.1071	0.9804	0.8234	0.1822	0.6193
RF	0.2411	0.9765	0.8443	0.3449	0.8272
ET	0.3571	0.9609	0.8523	0.4139	0.8651
CNN	0.4196	0.8493	0.7721	0.2587	0.7751
MLP	0.4196	0.9706	0.8716	0.5006	0.9048
LSTM	0.2857	0.9667	0.8443	0.3601	0.8025
GRU	0.3125	0.9393	0.8266	0.3142	0.8374

neoantigen peptides and 49 self-derived antigen peptides. Despite the limited sample size and the disease-specific nature of the data, NeoGen-BC demonstrated remarkably strong predictive performance on this breast cancer-focused test set (Table S9). The model correctly identified 24 of 25 neoantigens while misclassifying only a single self-antigen, resulting in a sensitivity of 0.9600, specificity of 0.9796, and an overall accuracy of 0.9730. Notably, the Matthews correlation coefficient (0.9396) and AUC (0.9951) further highlight the robustness and reliability of the predictions under this stringent evaluation.

These results provide compelling evidence that the immunogenic patterns learned from pan-cancer training data are transferable to breast cancer-specific antigen presentation contexts. The strong generalization observed across HR^+ and TNBC subtypes suggests that neoantigen immunogenicity may involve partially shared sequence-level and physicochemical characteristics across different cancer types. These findings support the possibility that deep learning models can capture conserved immunogenicity-associated features relevant to immune recognition, which may contribute to the development of scalable, off-the-shelf vaccines targeting recurrent breast cancer driver mutations.

3.5.2. Case study in ESR1 and PIK3CA mutations

To further investigate the biological relevance of NeoGen-BC in clinically actionable breast cancer drivers, we conducted a focused case study on shared neoantigens derived from recurrent ESR1 and PIK3CA mutations, including hotspot variants such as Y537S, D538G, and H1047R, curated from a recent experimental validation study of breast cancer neoantigens [26]. These mutations are particularly relevant in hormone receptor-positive breast cancer and are increasingly implicated in immune modulation under endocrine therapies.

Unlike conventional neoantigen pipelines that primarily emphasize MHC class I presentation, we specifically evaluated the ability of NeoGen-BC to identify MHC class II-presented neoantigens, which are critical for $CD4^+$ T cell activation, immune memory formation, and sustained antitumor responses. This focus is especially pertinent in the context of oral SERD-induced IFN-γ signaling, which has been shown to enhance antigen presentation and MHC-II expression in the tumor microenvironment, thereby creating a favorable setting for $CD4^+$ T cell-mediated immunity.

To benchmark NeoGen-BC against experimentally confirmed MHC-II presentation, we adopted the allele assignments and binding validation data reported by Ando et al. [26], who independently characterized these 7 ESR1 and 10 PIK3CA peptides using two orthogonal experimental methods: an MHC-density assay measuring cell-surface MHC-II–peptide complex formation by flow cytometry, and IFN-γ ELISPOT assays using peripheral blood mononuclear cells from HLA-typed healthy donors. Critically, rather than evaluating all peptides on a single allele, we assigned each peptide its experimentally confirmed restricting HLA class II allele based on a g15 ratio > 2.0, the threshold for strong MHC-II binding established by Ando et al. [26], and evaluated NetMHCIIpan 4.3 on the correct allele per peptide. This per-allele approach reflects the biological reality that different mutations within the same gene are presented by distinct HLA class II alleles across the population, including DRB4*01:03, DPA1*02:02-DPB1*05:01 (DP5), DRB1*04:05, DRB1*01:01, DRB1*15:01, and DRB1_15:02.

As summarized in Table S10, NeoGen-BC correctly identified 16 of 17 peptides as neoantigens. The single exception, C420R, lacked experimental support for both MHC-II binding (g15 ratio <2.0) and $CD4^+$ T cell recognition (ELISPOT penetrance < 0.4) in Ando et al.'s work, and was appropriately classified as non-neoantigen by NeoGen-BC, demonstrating that NeoGen-BC does not indiscriminately predict all mutation-bearing peptides as immunogenic.

When NetMHCIIpan 4.3 was evaluated on the experimentally confirmed restricting allele per peptide, it predicted binding (%Rank <10) for 13 of 17 peptides. However, four peptides remained discordant: K303R (%Rank 44.34), E380Q (%Rank 17.91), and G1049R (%Rank 11.72) on DRB4*01:03, and D538G (%Rank 36.78) on DP5. Notably, all four of these peptides were confirmed as strong MHC-II binders by the MHC-density assay (g15 ratios of 2.1, 2.3, 8.7, and 2.1, respectively) and elicited positive IFN-γ responses in $CD4^+$ T cell ELISPOT assays in Ando et al. [26]. This discordance is consistent with the explicit finding of Ando et al. that NetMHCIIpan 4.3 underestimates binding for specific allele-peptide combinations, likely due to sparse training data for certain HLA alleles, a limitation the authors confirmed by showing that the %Rank for E380Q worsened from 14.9 to 17.9 between NetMHCIIpan versions 4.1 and 4.3. G1049R represents the most striking case: despite a g15 ratio of 8.7, among the highest in the entire dataset, NetMHCIIpan assigns a %Rank of 11.72, placing it just outside the conventional binding threshold. NeoGen-BC correctly identifies all four of these peptides as neoantigens, recovering experimentally validated candidates that NetMHCIIpan fails to capture even on the correct restricting allele.

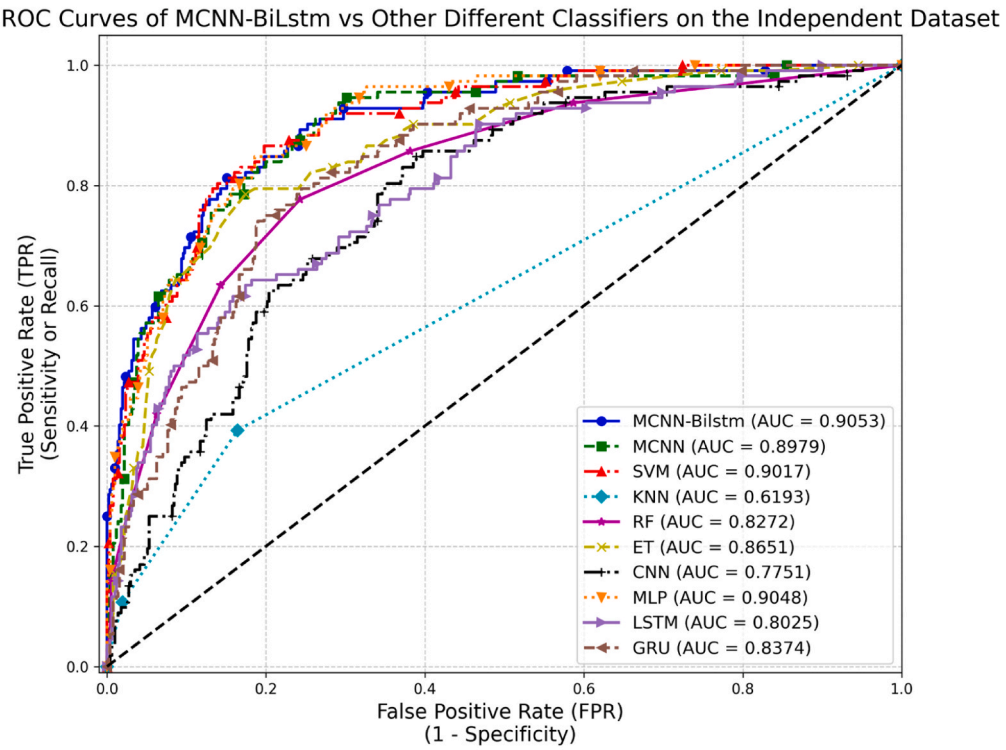
Interestingly, E726K showed a borderline g15 ratio of 1.5, below the strong-binder threshold of 2.0, yet exhibited a NetMHCIIpan %Rank of 1.78 on DRB4*01:03 and positive ELISPOT responses in Ando et al.'s work. This suggests that the g15 ratio threshold alone may not capture all immunogenic peptides, and that anchor motif analysis and T cell functional readouts provide complementary information. NeoGen-BC correctly identifies E726K as a neoantigen, consistent with the ELISPOT evidence.

In summary, this case study highlights important limitations of conventional MHC-II binding predictors. Single-allele evaluation is insufficient for heterogeneous shared neoantigens presented by diverse HLA-II alleles, and even allele-matched NetMHCIIpan 4.3 predictions failed to identify several experimentally validated binders reported by Ando et al. NeoGen-BC correctly identified 16 of 17 experimentally supported peptides, suggesting that immunogenicity-oriented features beyond binding affinity alone may improve neoantigen prediction and support the discovery of shared neoantigen candidates in HR + breast cancer.

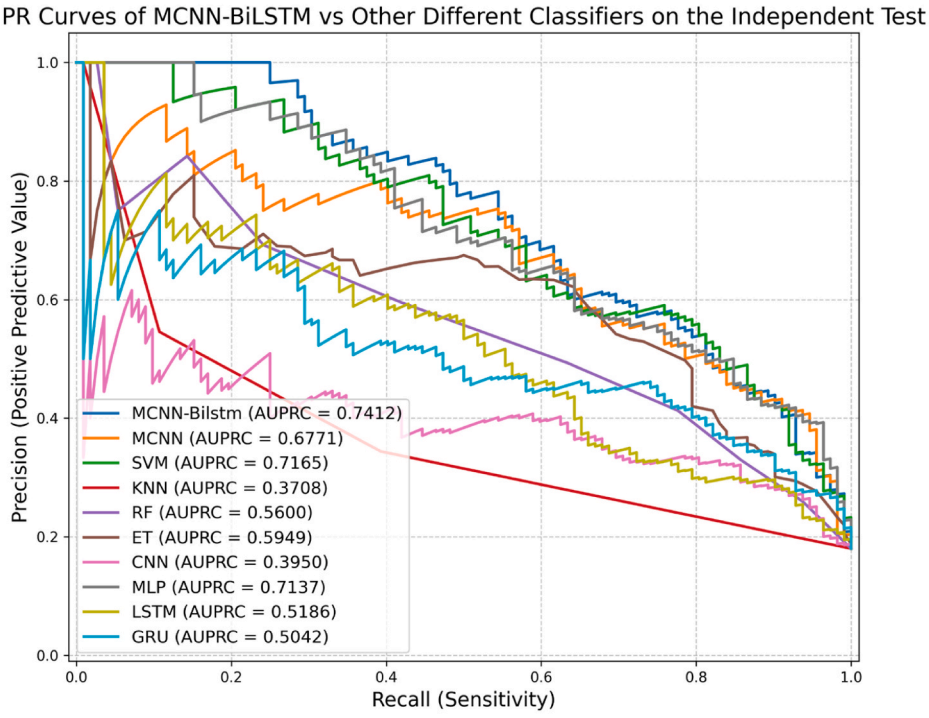
3.6. De novo generated peptides

In this section, we investigated whether AI-generated peptides share physicochemical and immunological properties with experimentally validated ESR1-derived neoantigens. This analysis aims to assess whether the generative model captures meaningful biological patterns rather than producing random amino acid sequences. We focused exclusively on ESR1 mutations, as ESR1 represents one of the most clinically relevant and recurrent driver genes in hormone receptor-positive breast cancer, with well-characterized neoantigens and established relevance to endocrine resistance mechanisms.

Seven recurrent ESR1 mutations reported by Ando et al. [26] were used as contextual prompts for the ProtGPT2 model, where each prompt consisted of a 15-residue sequence segment. Sequence generation was performed using the following decoding parameters: max_new_tokens = 20, temperature = 0.9, top_p = 0.95, top_k = 50, and repetition_penalty = 1.2, which together balance sequence diversity,



A



B

Fig. 4. Receiver Operating Characteristic (ROC) curves (A) and Precision–Recall (PR) curves (B) comparing the performance of different classification models with ProtT5-RAG features on the independent test set across varying classification thresholds.

biological plausibility, and redundancy control. For each mutation context, 1000 sequences were generated, yielding a total of 7000 de novo protein sequences.

To adapt these sequences for downstream immunogenicity analysis, a sliding-window approach [47] was applied to extract 15-mer peptides, resulting in 66,292 candidate peptides. After removing sequences containing non-amino acid characters, 66,135 peptides remained for further evaluation.

Given the stochastic nature of Transformer-based generation, not all generated sequences are expected to conform to realistic protein-like patterns. To prioritize high-quality peptides, we computed the perplexity score for each sequence, where lower perplexity indicates stronger conformity to the learned protein language model. The peptides were ranked by perplexity, and only the lowest one-third (22,045 peptides) were retained for subsequent analyses, ensuring a focus on sequences most consistent with natural protein statistics.

Although ProtGPT2 is trained on protein sequences, generated outputs may still deviate from fundamental protein organizational principles. To address this, we employed a geometric protein validation strategy based on the work of Yau et al. [49], which distinguishes true proteins from arbitrary amino acid sequences using statistical descriptors of amino acid distribution with 99.69% accuracy. Each peptide was represented by three quantities for each amino acid: the occurrence count (n_k), the mean positional distance (μ_k), and the second normalized central moment (D_2^k). Using the reviewed Swiss-Prot proteins from UniProtKB, 20 convex hulls were constructed, one for each amino acid, encompassing protein-like sequence space. Generated peptides were retained only if they fell inside all 20 convex hulls, resulting in 20,753 protein-like peptides.

To further evaluate the structural plausibility of candidate peptides, we applied ESMFold [38] to assess folding confidence for each remaining peptide. Unlike AlphaFold2, which relies on Multiple Sequence Alignments (MSAs) to infer evolutionary constraints, ESMFold leverages the internal representations of the ESM-2 protein language model to predict 3D structure directly from sequence, making it particularly efficient for short peptides where homolog searches are uninformative or unavailable [50]. We utilized the ESMFold model distributed via the Hugging Face platform [51] accessed through the Transformers library for flexible integration into our computational pipeline. For each isolated 15-mer peptide, ESMFold was used to generate predicted structures together with per-residue predicted local distance difference test (pLDDT) confidence scores. Importantly, pLDDT was employed in this phrase as a computational confidence and structural plausibility filter rather than as evidence of experimentally stable peptide conformations. Previous studies have shown that low pLDDT values are associated with increased structural disorder, conformational heterogeneity, and reduced confidence in predicted structures, particularly for random or de novo generated proteins [52]. Therefore, we used pLDDT to reduce the number of highly disordered or structurally uncertain generated peptide candidates prior to downstream filtering and peptide-MHC-II docking analyses. Specifically, peptides with mean pLDDT values greater than 70 were retained for further evaluation, following the commonly used high-confidence threshold reported in the original ESMFold study [38]. Applying this filter reduced the dataset to 1678 peptides with high-confidence predicted structures.

We next examined whether the AI-generated peptides exhibit physicochemical characteristics comparable to experimentally validated ESR1 neoantigens. Using the Python ProteinAnalysis module, we calculated hydrophobicity, aromaticity, molecular weight, net charge, isoelectric point, and instability index for all 1678 peptides. As shown in Fig. S3, the violin plot distributions of generated peptides substantially overlap with those of real ESR1-derived peptides, indicating that the generative model preserves key biochemical properties relevant to antigen presentation and immune recognition.

To further assess similarity in embedding space, we performed t-SNE analysis on real and generated ESR1 peptides (Fig. S4). To identify generated peptides most similar to experimentally validated ESR1 neoantigens, pairwise cosine distances were computed between generated peptide embeddings and ESR1 neoantigen embeddings within the reduced embedding space using the scikit-learn pairwise distance function. Peptides were ranked according to cosine distance, where smaller values indicate greater embedding similarity, and the top 25% closest peptides were retained, yielding 420 peptides that are most proximal to experimentally observed antigens. The selected peptides were subsequently evaluated using NetMHCIIpan 4.3 to assess MHC class II binding affinity. Peptides with strong (rank $< 2\%$) or weak (rank $< 10\%$) binding were retained, resulting in 208 peptides with favorable MHC-II binding profiles. These ranks follow the experimental criteria established by Ando et al. [26].

Finally, these peptides were assessed using our NeoGen-BC model to predict immunogenicity. All 208 selected peptides and 212 peptides which were discarded from the filtered 420 peptides are now assessed. Interestingly, the retained peptides consistently received higher prediction scores (0.65-0.71), whereas the discarded peptides predominantly received lower scores (0.35-0.45). These findings indicate that NeoGen-BC does not simply assign uniformly high immunogenicity scores to all generated peptides. Instead, the framework retains the ability to distinguish structurally and immunologically favorable candidates from lower-quality generated sequences. This behavior is consistent with the overall discriminative performance of NeoGen-BC observed in independent benchmark evaluations. This observation is illustrated in Fig. S5 using a violin-box plot comparing the NeoGen-BC prediction score distributions between retained and discarded generated peptides. Detailed binding affinities across 14 HLA-II alleles of these peptides are provided in our GitHub channel at https://github.com/s1129108/NeoGenBC.

Together, this multi-stage filtering and validation pipeline demonstrates that de novo peptides generated by ProtGPT2 not only resemble real ESR1 neoantigens in physicochemical and structural properties but also exhibit strong MHC-II binding and immunogenic potential. These results provide computational evidence that the generative component captures biologically relevant sequence and physicochemical characteristics associated with antigenicity, supporting its potential application in rational neoantigen discovery for breast cancer.

3.7. Structural validation of de novo peptides using AlphaFold-Multimer

To further substantiate the biological plausibility of the de novo-generated neoantigen peptides, we performed structure-level validation by explicitly modeling their interactions with human MHC class II (MHC-II) molecules using AlphaFold-Multimer [53]. While binding affinity predictions and immunogenicity scores provide valuable functional insights, structural compatibility between peptides and MHC-II binding grooves represents a critical prerequisite for antigen presentation and subsequent $CD4^+$ T-cell recognition.

For each selected de novo peptide, we constructed peptide-MHC-II complexes consisting of a fixed α chain (HLA-DRA*01:01) and one of six representative β chains: HLA-DRB1*01:01, HLA-DRB1*03:01, HLA-DRB1*04:01, HLA-DRB3*01:01, HLA-DRB4*01:03, and HLA-DRB5*01:01. This panel was chosen to reflect common HLA-DR alleles with broad population coverage and relevance to breast cancer immunology. Each peptide was modeled in complex with the corresponding MHC-II heterodimer, allowing AlphaFold-Multimer to jointly optimize the folding of the peptide and the MHC-II chains under a unified attention-based framework.

Fig. S6 presents representative AlphaFold-Multimer predictions, with residues colored according to pLDDT scores. Across multiple peptide-MHC-II complexes, the generated peptides consistently adopted extended conformations within the canonical MHC-II binding groove, stabilized by interactions with both the α-helical walls and β-sheet floor of the MHC-II molecule. Importantly, the majority of residues in both the

peptides and the MHC-II binding regions exhibited high pLDDT values (>90), indicating strong confidence in the predicted local structural conformations. In addition, the peptide-MHC-II complexes generally achieved high interface predicted Template Modeling (ipTM) scores (>0.9), suggesting strong confidence in the predicted intermolecular interfaces and relative positioning between the peptide and MHC-II chains. High ipTM values support the structural plausibility and stability of the predicted binding interactions rather than only the accuracy of individual chain conformations. Furthermore, the generated peptides aligned in orientations consistent with experimentally characterized MHC-II-restricted epitopes, collectively supporting their potential structural compatibility with antigen presentation machinery.

These structural observations provide an orthogonal layer of validation to our sequence-based and binding-based analyses. The ability of AlphaFold-Multimer to model stable peptide-MHC-II complexes suggests that the de novo peptides generated by our framework are not only predicted to bind MHC-II molecules but are also structurally capable of being presented to $CD4^+$ T cells.

In summary, these results underscore the end-to-end effectiveness of our approach, spanning sequence generation, biochemical validation, immunological prediction, and structural confirmation. This multi-level validation pipeline highlights the potential of our method to efficiently propose structurally viable neoantigen candidates for breast cancer peptide vaccine development.

4. Discussion

4.1. Protein language models can capture immunogenicity

Traditional neoantigen discovery pipelines have largely focused on predicting peptide-MHC binding affinity, operating under the implicit assumption that strong binders are more likely to be immunogenic. While MHC binding is a necessary condition for antigen presentation, it is increasingly recognized as insufficient to explain effective T-cell recognition and immune activation. Many peptides predicted as strong binders fail to elicit T-cell responses, whereas others with modest binding affinity can be highly immunogenic.

In this work, we demonstrate that protein language model (PLM)-based representations, particularly ProtT5 embeddings, encode rich contextual information that extends beyond binding motifs. These embeddings implicitly capture higher-order biochemical and evolutionary features such as residue co-dependencies, structural propensities, and sequence regularities that are directly relevant to T-cell receptor (TCR) recognition and antigen processing. The high performance of our MCNN-BiLSTM model, especially in distinguishing neoantigens from self-antigens across diverse datasets, supports the notion that PLMs function as immunogenicity encoders rather than mere binding predictors.

This distinction is further reinforced by our case studies involving ESR1 and PIK3CA mutations, where multiple experimentally validated $CD4^+$ T-cell epitopes were missed by NetMHCIIpan but correctly identified by our model. These results indicate that PLM-driven models can capture aspects of antigenicity that are orthogonal to canonical binding affinity scores, offering a more comprehensive framework for neoantigen prioritization.

4.2. Clinical relevance: shared neoantigens and support for the oral SERD-immunopeptidome strategy

A major clinical challenge in neoantigen-based immunotherapy is the heterogeneity of tumor mutations, which limits the feasibility of fully personalized vaccines. In this context, the identification of shared neoantigens, which are recurrent mutant peptides presented across patients, represents a critical opportunity to bridge precision medicine and scalable clinical implementation.

Our findings provide strong computational evidence that pan-cancer-trained models retain robust predictive power when applied to breast cancer-specific datasets, suggesting the presence of partially shared immunogenicity-associated features across different tumor contexts. In particular, the successful identification and structural validation of MHC-II-restricted neoantigens derived from ESR1 and PIK3CA hotspot mutations aligns closely with clinical strategies in which oral selective estrogen receptor degraders (SERDs) induce tumor remodeling, enhance interferon-γ signaling, and promote antigen processing and presentation.

Within this framework, the presentation of shared MHC-II neoantigens to $CD4^+$ T cells may act as a critical immunological amplifier, supporting both helper T-cell activation and downstream cytotoxic responses. By computationally prioritizing these shared peptides, our approach offers a rational foundation for next-generation breast cancer vaccines and combination therapies that integrate endocrine treatment with immunomodulation.

4.3. Limitations and future directions

Despite the encouraging results, several limitations should be acknowledged. First, the availability of experimentally validated neoantigen datasets remains limited, particularly for specific driver mutations and MHC-II-restricted epitopes. While our model demonstrates strong generalization across pan-cancer and breast cancer-specific cohorts, broader validation on larger and more diverse immunopeptidomics datasets will be essential to further establish its robustness.

Second, although structural modeling and in silico predictions provide strong evidence of antigen presentation potential, wet-lab validation remains indispensable. Future studies should incorporate functional assays such as Peripheral Blood Mononuclear Cell (PBMC)-based $CD4^+$ T-cell activation experiments, cytokine release assays, and tetramer staining to directly confirm immunogenicity. Such experimental validation will be particularly important for the de novo-generated peptides proposed in this study.

Finally, while our work emphasizes MHC-II-restricted neoantigens, immune responses are inherently multifactorial. Integrating additional layers such as antigen processing efficiency, TCR repertoire availability, and tumor microenvironment context represents a promising direction for further improving predictive accuracy.

5. Conclusion

In this study, we present NeoGen-BC, a comprehensive deep learning framework that extends beyond conventional neoantigen prediction to enable the rational design of next-generation breast cancer vaccines. By integrating protein language model embeddings, a multi-scale MCNN-BiLSTM classifier, retrieval-augmented learning, and a controlled generative module, NeoGen-BC captures fundamental principles of tumor immunogenicity rather than peptide-MHC binding alone.

A key strength of NeoGen-BC lies in its ability to identify and prioritize shared neoantigens derived from recurrent driver mutations, particularly ESR1 and PIK3CA, which represent common and clinically actionable vulnerabilities in breast cancer. Our results demonstrate that a model trained on pan-cancer data can generalize effectively to breast cancer-specific settings, suggesting the presence of partially conserved immunogenicity-associated features underlying immune recognition across different tumor contexts. Notably, NeoGen-BC successfully identifies MHC-II-restricted neoantigens that are frequently overlooked by traditional binding-based tools, highlighting its potential to uncover $CD4^+$ T-cell targets critical for durable anti-tumor immunity.

Beyond prediction, NeoGen-BC functions as a computational design platform. Through de novo peptide generation, stringent biophysical and structural filtering, and immunogenicity screening, we demonstrate an integrated in silico pipeline capable of prioritizing structurally plausible and potentially immunogenic neoantigen candidates. Structural evaluation using AlphaFold-Multimer further suggests that a subset

of these peptides may adopt conformations compatible with presentation within MHC-II complexes. Nevertheless, these generated peptides remain computational candidates and require experimental validation to confirm their immunogenicity, MHC presentation, and therapeutic potential.

Importantly, the identification of shared neoantigens from ESR1 and PIK3CA mutations is potentially relevant to emerging clinical strategies involving oral SERD-based therapies, which have been reported to influence tumor immune signaling and antigen presentation pathways. In this context, NeoGen-BC may serve as a computational framework for exploring possible connections between targeted therapy and immunotherapy by systematically prioritizing neoantigen candidates within therapy-associated immunogenic contexts.

In summary, NeoGen-BC represents a paradigm shift from static neoantigen prediction toward AI-driven vaccine design, enabling the discovery of shared, clinically relevant neoantigens in breast cancer. This framework lays the groundwork for future experimental validation and clinical translation, with the long-term goal of supporting scalable, mutation-informed peptide vaccines that complement modern breast cancer therapies.

CRediT authorship contribution statement

Van The Le: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization, Formal analysis. **Juan Peter Timothy Yuune:** Visualization, Validation, Data curation. **Jiun-I Lai:** Writing – review & editing, Investigation, Conceptualization. **Yu-Yen Ou:** Writing – review & editing, Supervision, Conceptualization, Funding acquisition, Project administration.

Data and software availability

Our code and dataset can be found on Github: https://github.com/s1129108/NeoGenBC.

Declaration of competing interest

I, Le Van The, hereby declare that I have no financial interests or relationships with any organizations that could potentially influence the subject matter of this work. I also confirm that I do not hold any professional or personal affiliations that may be perceived as affecting the impartiality and objectivity of my research.

I have received no funding, grants, or honoraria related to the research presented in this work. Additionally, I have no personal relationships or collaborations that might pose a conflict of interest. This work is conducted with complete transparency, and I am committed to upholding the highest standards of integrity in my scholarly contributions.

Acknowledgements

This work was partially supported by the National Science and Technology Council, Taiwan under Grant No. NSTC 112-2221-E-155-020-MY3 and NSTC 114-2221-E-155-028-MY3.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2026.154567.

References

[1] F. Carlino, C. Solinas, M. Orditura, M.D. Bisceglia, B. Pellegrino, A. Diana, Heterogeneity in breast cancer: clinical and therapeutic implications, Front. Oncol. 14 (2024) 1321654.
[2] X. Xiong, et al., Breast cancer: pathogenesis and treatments, Signal Transduct. Targeted Ther. 10 (1) (2025) 49.
[3] Q.H. Mohamed, E.I. Obeagu, Genetic heterogeneity in breast cancer: implications, Elite J. Health Sci. 2 (1) (2024) 20–24.
[4] Q. Maqsood, M.U. Khan, T. Fatima, S. Khalid, Z.I. Malik, Recent insights into breast cancer: molecular pathways, epigenetic regulation, and emerging targeted therapies, Breast Cancer Basic Clin. Res. 19 (2025) 11782234251355663.
[5] S.P.L. Galappaththi, et al., The genomic and biologic landscapes of breast cancer and racial differences, Int. J. Mol. Sci. 25 (23) (2024) 13165.
[6] C. Hao, Y. Wei, W. Meng, J. Zhang, X. Yang, PI3K/AKT/mTOR inhibitors for hormone receptor-positive advanced breast cancer, Cancer Treat Rev. 132 (2025) 102861.
[7] C. Tommasi, et al., Hormone receptor-positive/HER2-Positive Breast Cancer: hormone Therapy and Anti-HER2 treatment: an update on treatment strategies, J. Clin. Med. 13 (7) (2024) 1873.
[8] K. Jhaveri, F. Marmé, Current and emerging treatment approaches for hormone receptor-positive/human epidermal growth factor receptor 2-negative metastatic breast cancer, Cancer Treat Rev. 123 (2024) 102670.
[9] M.E. Tesch, et al., Estrogen levels in young women with hormone receptor-positive breast cancer on ovarian function suppression therapy, NPJ Breast Cancer 10 (1) (2024) 67.
[10] N. Xiong, H. Wu, Z. Yu, Advancements and challenges in triple-negative breast cancer: a comprehensive review of therapeutic and diagnostic strategies, Front. Oncol. 14 (2024) 1405491.
[11] V.B. Varzaru, T. Vlad, R. Popescu, C.S. Vlad, A.E. Moatar, I.M. Cobec, Triple-negative breast cancer: molecular particularities still a challenge, Diagnostics 14 (17) (2024) 1875.
[12] P. Schmid, et al., Overall survival with pembrolizumab in early-stage triple-negative breast cancer, N. Engl. J. Med. 391 (21) (2024) 1981–1991.
[13] C. Helal, et al., Clinico-pathological factors predicting pathological response in early triple-negative breast cancer, NPJ Breast Cancer 11 (1) (2025) 15.
[14] C. Hernando, et al., Oral selective estrogen receptor degraders (SERDs) as a novel breast cancer therapy: present and future from a clinical perspective, Int. J. Mol. Sci. 22 (15) (2021) 7812.
[15] T. Downton, F. Zhou, D. Segara, R. Jeselsohn, E. Lim, Oral selective estrogen receptor degraders (SERDs) in breast cancer: advances, challenges, and current status, Drug Des. Dev. Ther. (2022) 2933–2948.
[16] G. Guglielmi, M. Del Re, L.S. Gol, C. Bengala, R. Danesi, S. Fogli, Pharmacological insights on novel oral selective estrogen receptor degraders in breast cancer, Eur. J. Pharmacol. 969 (2024) 176424.
[17] Y. Lu, W. Liu, Selective estrogen receptor degraders (SERDs): a promising strategy for estrogen receptor positive endocrine-resistant breast cancer, J. Med. Chem. 63 (24) (2020) 15094–15114.
[18] T. Yan, L. Zhu, J. Chen, Current advances and challenges in CAR T-Cell therapy for solid tumors: tumor-associated antigens and the tumor microenvironment, Exp. Hematol. Oncol. 12 (1) (2023) 14.
[19] G. Dolton, et al., Targeting of multiple tumor-associated antigens by individual T cell receptors during successful cancer immunotherapy, Cell 186 (16) (2023) 3333–3349.
[20] M.M. Wolf, W.K. Rathmell, A.A. de Cubas, Immunogenicity in renal cell carcinoma: shifting focus to alternative sources of tumour-specific antigens, Nat. Rev. Nephrol. 19 (7) (2023) 440–450.
[21] W. Shi, Z. Tong, S. Chen, Q. Qiu, J. Zhou, H. Qian, Development of novel self-assembled vaccines based on tumour-specific antigenic peptide and TLR2 agonist for effective breast cancer immunotherapy via activating Cd8+ T cells and enhancing their function, Immunology 169 (4) (2023) 454–466.
[22] Y. Yan, P. Zhou, L. Ding, W. Hu, W. Chen, B. Su, T cell antigen recognition and discrimination by electrochemiluminescence imaging, Angew. Chem. Int. Ed. 62 (50) (2023) e202314588.
[23] E. Blass, P.A. Ott, Advances in the development of personalized neoantigen-based therapeutic cancer vaccines, Nat. Rev. Clin. Oncol. 18 (4) (2021) 215–229.
[24] P. Singh, et al., Advancements and challenges in personalized neoantigen-based cancer vaccines, Oncol. Rev. 19 (2025) 1541326.
[25] S. Ahmed, M.S. Mazhar, M.F. Shabbir, Neoantigen-based cancer vaccines: current innovations, challenges and future directions in personalized immunotherapy, Cancer Immunol Connect 1 (1) (2024) 1–10.
[26] Y. Ando, H. Miyadera, H. Bando, S. Hashimoto, E. Noguchi, H. Hara, HLA class II-restricted T cell epitopes in public neoantigens of ESR1 and PIK3CA in breast cancer, BMC Cancer 25 (1) (2025) 610.
[27] E. Kina, et al., Breast cancer immunopeptidomes contain numerous shared tumor antigens, J. Clin. Investig. 134 (1) (2024).
[28] A. Mauriello, B. Cavalluzzo, C. Ragone, M. Tagliamonte, L. Buonaguro, Shared neoantigens' atlas for off-the-shelf cancer vaccine development, J. Transl. Med. 23 (1) (2025) 558.
[29] N. Hashemi, et al., Improved prediction of MHC-peptide binding using protein language models, Frontiers in Bioinformatics 3 (2023) 1207380.
[30] X. Wang, et al., RPEMHC: improved prediction of MHC–peptide binding affinity by a deep learning approach based on residue–residue pair encoding, Bioinformatics 40 (1) (2024) btad785.
[31] R. You, W. Qu, H. Mamitsuka, S. Zhu, DeepMHCII: a novel binding core-aware deep interaction model for accurate MHC-II peptide binding affinity prediction, Bioinformatics 38 (Supplement_1) (2022) i220–i228.
[32] Y. Yu, et al., Structure-aware deep model for MHC-II peptide binding affinity prediction, BMC Genom. 25 (1) (2024) 127.
[33] A. Motmaen, J. Dauparas, M. Baek, M.H. Abedi, D. Baker, P. Bradley, Peptide-binding specificity prediction using fine-tuned protein structure prediction networks, Proc. Natl. Acad. Sci. 120 (9) (2023) e2216697120.

[34] J. Cheng, K. Bendjama, K. Rittner, B. Malone, BERTMHC: improved MHC–peptide class II interaction prediction with transformer and multiple instance learning, Bioinformatics 37 (22) (2021) 4172–4179.
[35] Claus Lundegaard, Ole Lund, Søren Buus, Morten Nielsen, Major histocompatibility complex class I binding predictions as a tool in epitope discovery, Immunology 130 (3) (2010) 309–318, https://doi.org/10.1111/j.1365-2567.2010.03300.x. https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2567.2010.03300.x. (Accessed 9 June 2010).
[36] C.-C. Chuang, Y.-C. Liu, Y.-Y. Ou, DeepNeoAG: neoantigen epitope prediction from melanoma antigens using a synergistic deep learning model combining protein language models and multi-window scanning convolutional neural networks, Int. J. Biol. Macromol. 281 (2024) 136252.
[37] A. Elnaggar, et al., Prottrans: toward understanding the language of life through self-supervised learning, IEEE Trans. Pattern Anal. Mach. Intell. 44 (10) (2021) 7112–7127.
[38] Z. Lin, et al., Evolutionary-scale prediction of atomic-level protein structure with a language model, Science 379 (6637) (2023) 1123–1130.
[39] R. Rao, et al., Evaluating protein transfer learning with TAPE, Adv. Neural Inf. Process. Syst. 32 (2019).
[40] P. Lewis, et al., Retrieval-augmented generation for knowledge-intensive nlp tasks, Adv. Neural Inf. Process. Syst. 33 (2020) 9459–9474.
[41] Z. Jiang, et al., Active Retrieval Augmented Generation, 2023, pp. 7969–7992.
[42] H. Yu, A. Gan, K. Zhang, S. Tong, Q. Liu, Z. Liu, Evaluation of retrieval-augmented Generation: a Survey, Springer, 2024, pp. 102–120.
[43] V.-T. Le, M.-S. Malik, Y.-J. Lin, Y.-C. Liu, Y.-Y. Chang, Y.-Y. Ou, ATP_mCNN: predicting ATP binding sites through pretrained language models and multi-window neural networks, Comput. Biol. Med. 185 (2025) 109541.
[44] V.T. Le, et al., NAD_MCNN: combining protein Language models and multiwindow convolutional neural Networks for Deacetylase NAD+ binding site prediction, Chem. Biol. Drug Des. 105 (4) (2025) e70085.
[45] V.T. Le, J.P.T. Yuune, T.T.P. Vu, M.S. Malik, Y.-Y. Ou, DeepCR: predicting cytokine receptor proteins through pretrained language models and deep learning networks, J. Biomol. Struct. Dyn. (2025) 1–18.
[46] N. Ferruz, S. Schmidt, B. Höcker, ProtGPT2 is a deep unsupervised language model for protein design, Nat. Commun. 13 (1) (2022) 4348.
[47] I. Malaina, et al., A universal antigen-ranking method to design personalized vaccines targeting neoantigens against Melanoma, Life 13 (1) (2023) 155.
[48] W. Li, A. Godzik, Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences, Bioinformatics 22 (13) (2006) 1658–1659.
[49] S.S.T. Yau, X. Zhao, K. Tian, H. Yu, Convex hull principle and distinguishing proteins from arbitrary amino acid sequences, in: Mathematical Principles in Bioinformatics, Springer, 2023, pp. 115–138.
[50] S.K. Mahtha, S. Venkadesan, D. Mohanty, Comparative evaluation of the prediction accuracy of AlphaFold and ESMFold for monomeric and dimeric proteins, NAR Genom. Bioinf. 8 (1) (2026) lqag002.
[51] S.M. Jain, Hugging face, in: Introduction to Transformers for NLP: with the Hugging Face Library and Models to Solve Problems, Springer, 2022, pp. 51–67.
[52] L. Middendorf, L.A. Eicholt, Random, de novo, and conserved proteins: how structure and disorder predictors perform differently, Proteins: Struct., Funct., Bioinf. 92 (6) (2024) 757–767.
[53] J. Jumper, et al., Highly accurate protein structure prediction with AlphaFold, nature 596 (7873) (2021) 583–589.